

Network Documentation — Guided Notes ANSWER KEY

Unit 3.1.1 | CompTIA Network+ N10-009 Obj. 3.1 | N10-008 3.1

For instructor use / distribute after students complete the notes.

Question 1

A _____ diagram shows the logical layout of a network — IP address ranges, VLANs, routing paths, and subnets — without regard to physical location. A _____ diagram shows where devices actually sit — buildings, racks, cable runs, and port connections.

Answer: Logical diagram; physical diagram.

Question 2

What is IPAM, and why is it more reliable than a spreadsheet for tracking IP address assignments on a large network?

Answer: IPAM (IP Address Management) is a dedicated system that tracks IP address allocations, DHCP scopes, and DNS records in one place. Unlike a spreadsheet, IPAM enforces uniqueness, flags conflicts, integrates with DHCP and DNS servers, and provides search and audit capabilities. Spreadsheets go stale and create conflicts; IPAM stays authoritative.

Question 3

A VLAN table documents each VLAN's _____, _____, and the _____ it connects to. Without this table, adding a new switch to the network means guessing which VLANs to configure.

Answer: VLAN ID, name/purpose, and the subnets (or SVIs/default gateways) it connects to.

Question 4

A network baseline captures normal metrics — bandwidth utilization, CPU load, latency — during _____ operation. Explain how a baseline helps during troubleshooting.

Answer: Normal (steady-state) operation. A baseline gives you a reference point — if current latency is 40ms and the baseline is 2ms, something is wrong. Without a baseline, you can't distinguish abnormal from normal. It turns 'this feels slow' into 'this is 20x above normal.'

Question 5

Rack documentation should include each device's _____, its _____ connections (what's plugged into which port), and the device's _____. Why does this matter when a device needs to be replaced at 3 AM?

Answer: Rack unit (RU) position; cable/port connections; management IP (and console access method). At 3 AM, you need to know where the device is, how to access it, and what it connects to — without turning every cable over to read labels or checking a colleague's memory.

Question 6

The _____ log records every configuration change made to the network — what changed, when, who made it, and why. How does this directly speed up troubleshooting?

Answer: Change log. When something breaks, the first question is 'what changed?' A change log answers that in seconds. Without it, troubleshooting starts with hypothesis and elimination instead of a known starting point.

Question 7

Standard Operating Procedures (SOPs) document how to perform _____ tasks. Explain why 'knowledge in people's heads' is a liability, and what happens when those people are unavailable.

Answer: Routine/repeatable tasks. Undocumented knowledge creates a single point of failure in the team. When the person who knows the process is on vacation, sick, or leaves the organization, that knowledge walks out with them. SOPs make processes reproducible regardless of who performs them.

Question 8 — Real World Application

REAL WORLD: You inherit a network with no documentation. Describe the order in which you would build documentation from scratch, and explain why you would start with that first item.

Answer: Start with a physical diagram and device inventory — you need to know what exists and where before you can document anything else. Then: logical diagram (Layer 3 topology, subnets), VLAN table, IP address inventory (IPAM or spreadsheet), cable labels, then SOPs. The physical inventory comes first because everything else builds on knowing what devices are present.